

Bayesian Batting Average Estimation: A Beta-Binomial Approach to Small-Sample Performance

Abstract:

Introduction:

The normal metric for calculating batting average in baseball, in this paper referred to as a *naïve average*, is found using the formula:

$$AVG = \frac{hits}{at\ bats}$$

That is: the ratio of the number of times a player made contact with the ball to the number of chances they had. However, this particular statistic ignores sample size. For example, a player going 6/10 and 600/1000 both have a naïve average of 0.600, but they are not equally convincing. A player going 6/10 is a good weekend of baseball, one going for 600/1000 is the best hitting season of all time.

Oftentimes, earlier in the season or in leagues where fewer games are played (such as the U.K. baseball leagues) naïve average is used as justification for a player's hitting ability when the sample size is far too low to make any serious judgment, i.e. 'he had a good weekend'. So, this paper considers a better statistic for a player's true hitting ability in these cases, attempting to answer the question: "*How does Bayesian estimation handle small-sampling batting averages compared to naïve batting average*".

Mathematical Background:

This experiment aimed to produce a better estimate of a player's true batting ability, that being their true probability of getting a hit, which is labelled as p . As mentioned, the traditional naïve average, $\hat{p}$, is calculated via:

$$\hat{p} = \frac{h}{n}$$

p = true probability of a hit, $\hat{p}$ = naïve batting average, h = observed hits, n = observed at bats

So, Naïve batting average is an attempt to estimate $\hat{p}$, it is not the same thing as p . Clearly, a better calculation is necessary as $\hat{p}$, makes no consideration of sample size. Again:

$$\frac{6}{10} = 0.600$$

$$\frac{600}{1000} = 0.600$$

These have the same observed average, but hugely different amounts of evidence. So, it is untrue to argue that both of these hitters have the same ability, a method considering sample size and uncertainty must be used.

At-Bats were assumed to be independent of each other and that the same probability of getting a hit was present each plate appearance. Therefore, given the binary nature of an at-bat (that being a hitter either hits the ball or gets out), at-bats were modelled as a Bernoulli Trial:

$$X \sim \text{Bernoulli}(p)$$

Where:

$$X = 1 \text{ means a hit}$$

$$X = 0 \text{ means an out}$$

Now, given the assumptions provided, if a player has true hit probability p , then the number of hits over n at-bats can be modelled using a Binomial distribution:

$$H | p \sim \text{Binomial}(n, p)$$

$$H = \text{number of 'new' hits achieved, } n = \text{number of 'new' at bats}$$

Bayesian statistics fits our new constraints as it represents uncertainty over p using a distribution. It allows us to use our pre-known data, the prior, to calculate the updated p given new data. Bayes' Theorem:

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)}$$

$$\therefore P(p | \text{data}) \propto P(\text{data} | p)P(p)$$

Hence, the posterior combines what was believed before seeing the new data with how well different values of p explain the observed data. Making it a much stronger estimation for a true prediction of a player's hitting ability.

Since p is a probability which lies from $0 \leq p \leq 1$, a Beta distribution is used as a natural prior for the unknown probability:

$$p \sim \text{Beta}(\alpha, \beta)$$

$$\alpha = \text{observed previous hits, } \beta = \text{observed prior outs}$$

That being, the model used for previously held information on a player's hitting, upon which we add the new data. The mean for a Beta distribution is calculated using:

$$E[p] = \frac{\alpha}{\alpha + \beta}$$

$$(\alpha + \beta) = \text{hits} + \text{outs} = \text{total at-bats}$$

This experiment used a simple prior of

$$Beta(25,75)$$

That being one centred around 0.250, a common MLB average, which acts as 100 prior pseudo-at-bats.

So, when trying to update our estimate for p , the update rule adds newly observed hits to α and newly observed outs to β .

$$p \mid h, n \sim Beta(\alpha + h, \beta + n - h)$$

$$h = \text{newly observed hits}, (n-h) = \text{newly observed outs}$$

Ultimately, the estimate we will actually use will be our posterior mean, the most accurate single number answer we can give of the Bayesian estimate. It lies between the prior expectation and the observed average. Small samples stay close to the prior, large samples move closer to the observed average. This pulling toward the prior is called shrinkage.

$$E[p \mid h, n] = \frac{\alpha + h}{\alpha + \beta + h}$$

Moreover, we can further the accuracy of our estimate by measuring uncertainty. The posterior mean gives a single number which is useful for initial observation; however, given the posterior is a distribution it also gives a range of plausible values. This experiment used a 95% credible interval, that being the interval in which the central 95% of posterior probability is contained. Overall, larger samples should produce narrower credible intervals because the model has more evidence i.e. is more confident.

Method:

This experiment utilised Python to for calculations, accessible at:

src/bayesian_batting.py

The key function's written were those to calculate naïve average, posterior parameters, the posterior mean and the credible interval. Throughout this experiment the prior mentioned above was used for all examples:

$$Beta(25,75)$$

This was chosen as it is centred at 0.250 and behaves like 100 prior at-bats, both reasonable priors for all cases tested.

There was also a function *batting_summary* used which calculated all outputs required for one input/player. Using this function, the same batting average of 0.600 was tested across increasing sample sizes, those being:

Index	Hits	At-Bats
0	3	5
1	6	10
2	30	50
3	60	100
4	300	500

Figure 1 – A table illustrating the cases tested

Results:

From the cases tested, the following table was produced, displaying the results of each of the statistics measured in each case:

Index	Hits	At bats	Naïve average	Posterior mean	Posterior Standard Deviation	Credible Interval Lower	Credible Interval Upper	Credible Interval Width
0	3	5	0.6	0.259	0.043	0.181	0.346	0.165
1	6	10	0.6	0.267	0.043	0.191	0.351	0.160
2	30	50	0.6	0.306	0.043	0.241	0.375	0.134
3	60	100	0.6	0.327	0.043	0.271	0.385	0.114
4	300	500	0.6	0.361	0.043	0.330	0.393	0.063

Figure 2 – Table displaying all calculations for each of the cases tested

Whilst all of the data is displayed here, graphs were plotted using this information to make the analysis more intuitive.

Firstly, a graph was plotted to display change over number of at bats for both Naïve batting average and the new Bayesian estimate:

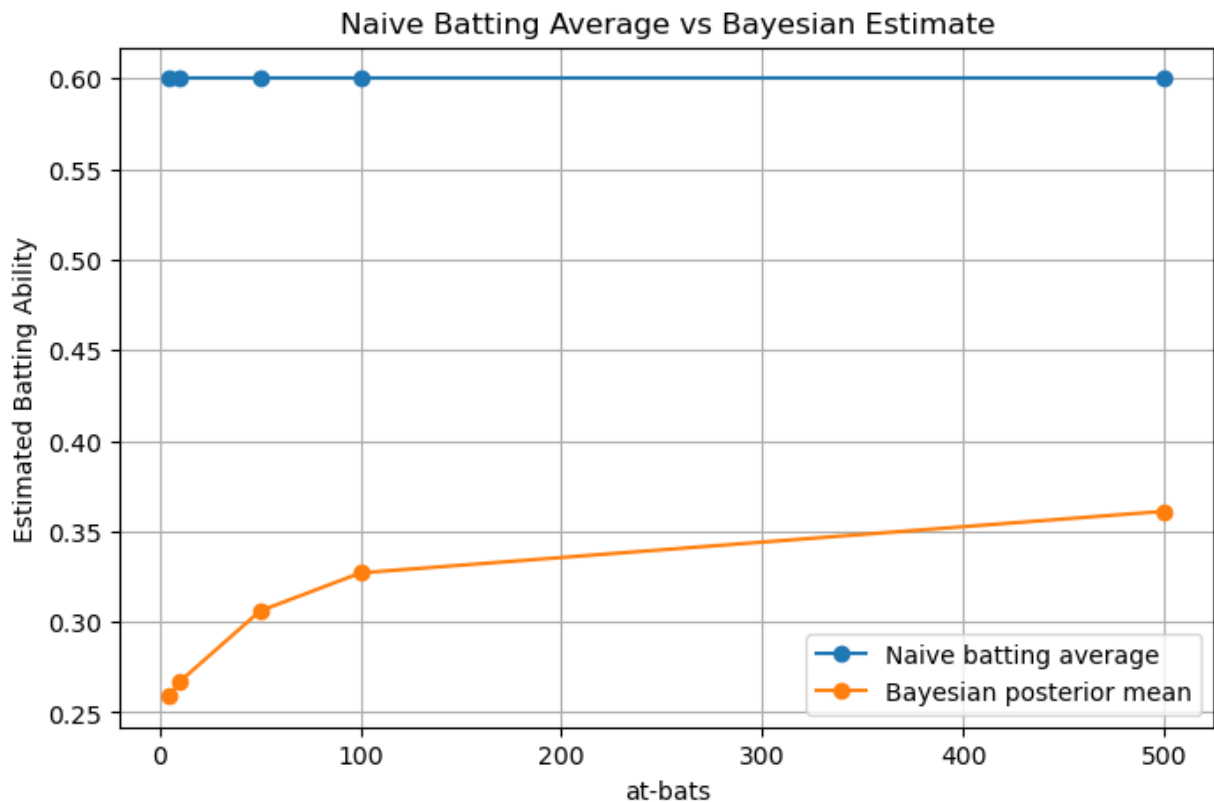


Figure 3 – A graph showing naïve batting average vs the Bayesian estimate

From which, it can be observed that, whilst the naïve batting average remains at 0.600 regardless of the number of at-bats taken, the Bayesian estimate starts much lower and moves toward 0.600 as the sample size increases. That is, as the confidence in $p=0.600$ increases, the Bayesian estimate moves towards such a value.